ALY 6980 Capstone - Winter 2026

Instructor: Valerie Atherley

Module - 6 Assignment: Individual Project Proposal Analysis and Edits

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Submitted by,

Sumesh Chakkaravarthi Purushothaman

Overview and Rationale

The Summit Global Consulting project aims to create a unified decision-making system using multiple vendor and partner data sources by transforming their separate data into a single strategic decision support system for business operations. The initial business problem which involves merging multiple spreadsheets into one standard system requires operational solutions but the actual work for the project requires them to solve data governance problems and maintain data accuracy while using ethical analytics methods to assess their healthcare service procurement needs and their partnerships with healthcare providers.

As healthcare organizations increasingly work with third-party vendors to deliver their clinical and operational functions many organizations experience difficulties because they cannot access identical vendor data which exists in the United States through various vendor-specific sources which present vendor data in different formats and assess service quality to vendors through compliance-based methods instead of value-based methods. The healthcare executive team faces challenges which make it difficult to reach fair and responsible vendor use decisions because of vendor-related data assessment difficulties. Summit Global Consulting enables organizations to gather all vendor data into one database which allows them to exit their present method of gathering vendor data without using it for ongoing vendor assessment. The organization will have an active and structured approach to create essential information about vendor data which enables them to establish logical and defendable vendor management decisions through their vendor assessment process which includes both supplier diversity and operational efficiency evaluations.

Purpose:

The goal of this initiative is to shorten the time to get products to the market by helping Summit Global Consulting locate and rank high-value IT-industry vendors that 1) demonstrate a committed supplier diversity and that 2) can be verified to have current, active operations and that 3) can be easily contacted.

This will be accomplished by moving from passive reporting to structured vendor intelligence and converting raw data into reasoned decisions.

Audience:

This system is designed specifically for business development managers, procurement analysts, supplier diversity officers and executive leadership. These groups will expect high-confidence analytics that can be easily understood, can be defended and align with their strategic outreach objectives. Therefore, this system has been developed for non-technical business professionals rather than data scientists.

Scope:

This initiative is purposefully limited to:

- IT-sector vendors
- validated vendor contact information
- active vendor operations

- top 20-50 ranked vendors based upon various criteria.

Also, predictive modelling and external benchmarking will not be used so as to avoid bias and maintain transparency in the governance and methodology employed.

Intersection with Data Governance, Leadership, Ethics, and Social

Responsibility

Data Governance

The project handles data governance through multiple aspects which manage large amounts of both structured data and unstructured data through established data standards and data transparency procedures and data accountability measures. Vendor datasets currently lack standard schema which creates challenges for measuring vendor performance and assessing vendor diversity commitment and evaluating vendor alignment with strategic objectives. The organization needs to establish a standard definition of data which will enable it to regulate data classification because this process establishes crucial data integrity and traceability requirements that support effective data governance in healthcare analytics.

Leadership from Business

The solution focuses on strategic vendor selection in the decision-making process instead of managing supplier activity based on reactive behaviours. Adobor and McMullen (2014) state that supplier diversity initiatives will create value when leadership implements performance measurement as a tool to evaluate vendors who can become strategic partners while supporting supply chain resilience through resource allocation.

Corporate Social Responsibility and Ethical

The Project Analysis section of this analysis requires individuals to explain their actions through direct evidence which enables physicians and administrators to assess results from both analytical analysis and output. According to Obermeyer and Emanuel (2016), clinical analytics should depend on human judgment and clinical experience because these methods will prevent all forms of bias and data misuse. The project will include Supplier Development Organization (SDO) Commitment metrics in each of the previous analyses to establish and demonstrate responsible procurement practices pertaining to and support the definition of supplier diversity initiatives as non-symbolic but rather as defined compliance and performance metrics.

Literature Review

The rationale for the development of this project derives from five strands of literature:

1. **Adobor & McMullen (2014):**

Companies, based on past studies, provide greater business value by including a Supplier Diversity program (i.e., having diversity for their vendors over and above whether or not they are able to

comply with databases or through the creation of their own internal metrics). In doing so, they build their flexibility to adapt quickly to changes in the marketplace with a diverse set of suppliers based on standard processes and policies used for assessing all suppliers. Supplier Diversity as a function should be used from a strategic asset perspective for firms whose vendors include complex networks of businesses within a single industry grouping.

The research provided by Adobor and McMullen also demonstrates the relationship between developing a methodology for rankings and categorizing supplier firms in addition to focusing on the supplier's performance based on established metrics (either uniformly or per category) within the scope of developing a comprehensive dashboard solution from Summit Global Consulting to provide an empirical, analytical decision support tool for rapidly assessing the operating performance of current and potential diverse supplier relationships rather than simply providing a report on the vendor's diverse supply base via the use of SDO metrics.

2. Whitfield & Landeros (2006):

In their study, Whitfield and Landeros (2006) stated that large organizations have many obstacles blocking small and diverse suppliers from entering their supply chains. The authors further identified that large organizations encounter three barriers to developing supplier inclusiveness in their supply chain: information asymmetry; lack of transparent vendor directories; and difficulty in identifying proper organizational contact points for small and diverse suppliers. They concluded that having centralized and well-organized vendor information systems would greatly enhance a supplier's ability to access a purchasing organization by offering more qualified vendors from which to select when making purchases.

This study provides validation for the dashboard-centric model of analytics that is being proposed in this project to support timely supplier decision-making. The sponsor indicated that delays in purchasing arise due to inadequate/outdated contact information and non-active vendors. By creating an analytics-based supplier inclusion governance framework that necessitates evidence of both current contact information and current business status to be included in the analytics database, the proposed analytics platform will address the aforementioned transparency issues.The literature therefore reinforces the need for governance-driven data curation versus the unrestricted accumulation of data.

3. Obermeyer & Emanuel (2016):

The use of machine learning and predictive analytics in healthcare has expanded quickly according to Obermeyer and Emanuel 2016 yet predictive model systems need proper controls and understanding methods before their safe implementation. The authors acknowledge that advanced analytics provide benefits yet they assert that all healthcare analytics decisions must remain unbiased and follow ethical standards and specific domain requirements while ensuring complete openness. Human judgment should receive support from analytics because this situation applies to decisions in high-pressure situations.

The authors' findings provided insight into how Summit Global Consulting utilized the predictive analytics approach in their proposal; Summit Global Consulting uses predictive models while they present high-confidence analytics that users can verify through dashboard-based analytics and decision support systems. The solution maintains its ethical analytics protection through two established processes which generate interpretable results from predictive analysis methods while following ethical standards that require auditability about leadership decisions which need accountability, and the solution meets Obermeyer and Emanuel's definition of ethical predictive analytics usage because it provides decision support through predictive analytics instead of predictive analytics itself.

4. Khatri & Brown (2010):

Khatri & Brown (2010), examine the importance of "data governance" for businesses, especially in industries where there are complicated regulations and operations. Khatri and Brown (2010), define "data governance," as an organization-wide framework containing the right processes, practices, and people to deliver high quality, reliable, and accountable data. Khatri and Brown's (2010) research highlights the importance of governance structures; without effective governance structures, the organization's analytical outputs will likely not be reliable or useful, regardless of technology used.

The proposed analytics platform supports data governance principles by using standardized schemas, controlled categorization of data, and using only reliable data sources. The sponsor required that vendors who provided unverifiable contact information not have access to the analytics platform as it is consistent with Khatri & Brown's (2010) assertion that data trustworthiness must take priority over data completeness in governance decisions. Therefore, the objective of the project (to create maintained datasets that support the executive team's ability to make informed and

defensible decisions) will support the executive team's ability to make confident and defensible decisions.

5. Davenport & Harris (2007):

The study conducted by Davenport and Harris (2007) investigates various stages of analytics maturity which organizations need to develop analytics capabilities. The authors present five stages of analytics maturity which organizations can use to assess their current analytics capabilities that range from basic reporting functions to using analytics for strategic market advantages. The authors argue that organizations can achieve successful analytics implementation only if they integrate analytics solutions into their existing operational processes and leadership structures. According to the value of analytics becomes accessible to businesses when organizations obtain operational analytics that match their business requirements.

The dashboard design philosophy of Summit Global Consulting base itself on this framework which serves as its main foundation. The sponsor clearly discussed that simplicity and usability will be important for stakeholders who do not have a technical background in making decisions regarding Vendor Partner Lists. The organization selected to develop a Vendor Partner Lists dashboard because they wanted to focus on this feature instead of creating complex models which would evaluate supplier data through various product lines and advanced analytic techniques. The recommended solution provides analytics in an accessible format which helps users who need to make decisions about partnerships and outreach, while simultaneously functioning as a strategic resource for organizational leadership.

Data Points for Proposal

The following data points will be included in this proposal:

- Vendor ID (name, contact information, location)
- Industry/Sector (e.g., healthcare, transportation, IT, facilities)
- Service Code (e.g., MEDICAL, VEHICLE, FACILITY)
- Supplier Diversity Office (SDO) commitment level (%)
- Contract type and current status of engagement
- Performance measures (responsiveness, service area, service delivery rate)
- Geographic area where services are provided by provider/contractor.

Normalization and validation of all data elements will take place before being included in the analytical model (e.g., percentage-based variables such as SDO commitment will be standardised to

enable weighted comparisons with financial indicators). To ensure that all vendors included in the ranking system are operationally accessible, we will use contact verification as a governance control variable. This will enhance both comparability and facilitate transparent decision making through the systematic transformation of raw data into analytical variables.

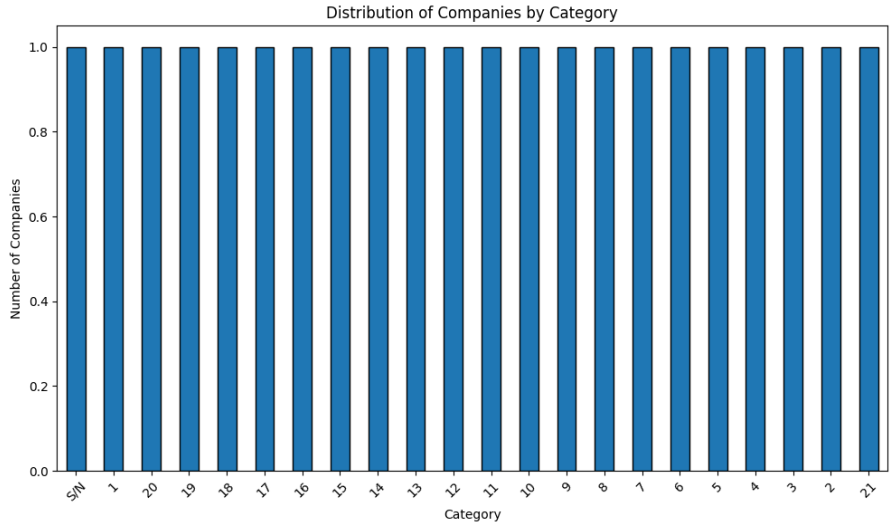
Sources of Data

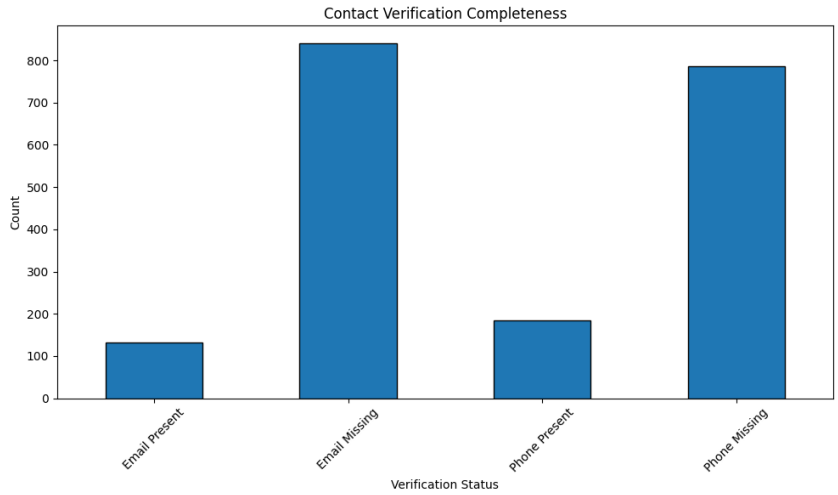
Summit Global Consulting uses internal vendor spreadsheets as one of its main data collection methods. Summit Global Consulting will combine internal vendor spreadsheets with state-provided supplier diversity information. The organization will extend its internal data collection by acquiring public procurement and contract information from third-party data sources which include USAspending. Gov (federal) and the State of Massachusetts' procurement portal (for IT and professional services classifications).

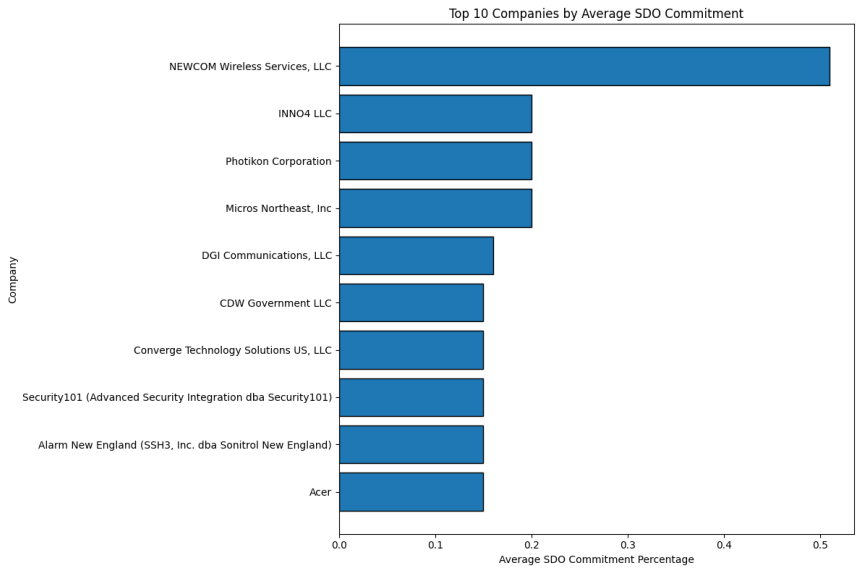
The team will use publicly available vendor websites and records from the Supplier Diversity Office to improve their knowledge about vendor activities and to gather up-to-date contact details for every vendor. The final consultant report will exclude vendors who lack verified contact information or physical presence according to the evaluation conducted by the sponsoring organization.

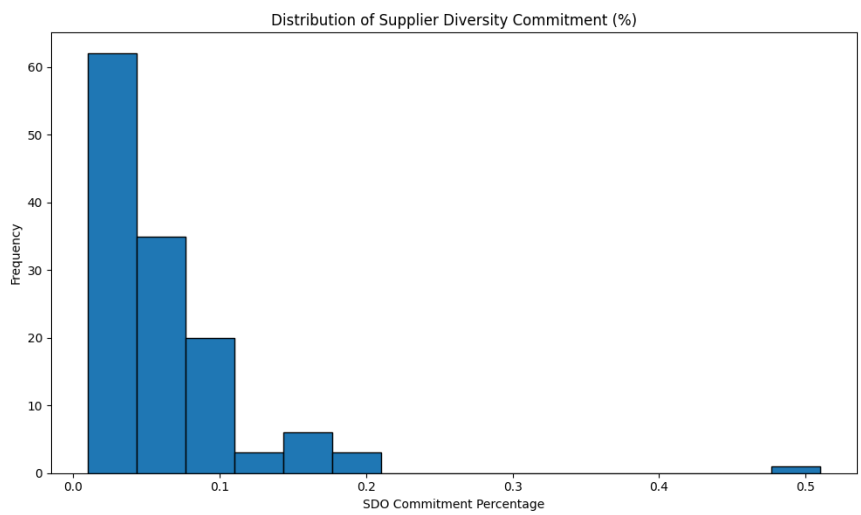
Exploratory Data Analysis

Before the ranking model is developed, an exploration of the key vendor parameters will be completed with EDA in order to better understand their distribution and their specific behaviours. Some analysis will include examining the distribution of Commitment Percentage SDOs, the correlation between estimated addressable spend versus the diversity commitments made by vendors, the number of inactive vendors per region, and when vendors are located in a clustered manner to one another in the IT vendor space. The outcomes of the EDA part of this process will also assist with weight calibration for the ranking model and will identify participants that could be considered outliers, or will expose dataset structural biases (e.g. too strong of an emphasis on a particular factor). By ensuring a basis for RFQ ranking logic is created based on observed descriptive data patterns, the ranking system will rely on sound governance and will reduce the likelihood of arbitrary determinations being made during the prioritization of each RFQ.









While certain financial and contract variables are referenced for future model expansion, the current exploratory analysis focuses on diversity commitment percentages and contact verification completeness available within the structured datasets.

Internal Stakeholder Validation Strategy

An external survey is not currently going to be conducted. However, we may create a simple internal stakeholder survey to collect qualitative input on perceived reliability of vendors or importance to Summit Global Consulting's strategy. These responses would be accumulated for validation purposes rather than predictive modelling.

To measure the perceived reliability of the data being used in this process, assess the importance of the diversity metrics that are used and evaluate buy-in on the use of weighted vendor prioritization, business development and procurement staff will complete a Stakeholder Validation Survey. The survey will include both Likert-based and open-ended questions to gather quantitative and qualitative input. Quantitative data will be analyzed using descriptive statistics, including mean and frequency distributions; qualitative data will undergo thematic analysis to uncover governance issues and potential adjustments to weighting elements. The conclusions reached by the analysis will inform recommendations for modifying ranking weights and enhancing dashboard usability to ensure that the analytical framework meets the expectations of Senior Leadership.

Approach to Analytics; Dashboarding vs. Predictive Modelling

Dashboarding will provide better decision support according to project stakeholders than predictive models. The sponsor indicated that the primary challenge was the inability to quickly identify active IT-sector companies that have a significant commitment to Supplier Diversity Office (SDO) spend (in the form of diversity spend) and to identify valid types of contacts. The organization no longer requires predictive analytics because it needs to focus on outreach activities while making priorities for its operations.

The dashboard will display high confidence data fields which include the company's status and SDO commitment percentage and estimated addressable spend and verified contact details. The principles establish ethical analytics practices which eliminate speculative inferences while delivering data representations that enable leadership decision making through transparent auditable source access.

VENDOR SCORING MODEL

The Vendor Prioritization Framework will utilize a weighted scoring approach to assess vendors by incorporating their commitment to diversity, financial stability, operational readiness, and contact verification. The weighting is as follows: 40% SDO Commitment Percentage; 30% Normalized Addressable Spend; 20% Active Contract Status; and 10% Contact Verification Status. This weighting scheme reflects how the Sponsors prioritize operationalizing diversity while giving equal importance to both outreach feasibility and financial alignment. These weighting elements will remain flexible and will continue to be refined through stakeholder validation feedback.

Financial and contract-based weights represent planned future data integration once structured procurement datasets are fully incorporated.

Sponsor Validation and Alignment

The sponsor approved all aspects of the project scope including its complete description and the analysis methods which confirmed that end users would operate through direct contact with project stakeholders. The sponsor established three requirements for project development which required non-technical business users to access accurate content through simple design elements; this decision created a project focus on dashboard development and vendor ranking system while preventing any work on data infrastructure and predictive analytics development.

Example Use Case

A Business Development Manager at Summit Global Consulting opens up the dashboard to find IT sector companies that are spending the most money on supplier diversity programs (as a percentage of total annual spend).The results of this search will provide information about the companies' commitment to SDO, estimated addressable spend, and verified contact information for the Supply Diversity Managers at these companies in just seconds.

With this data, the user can quickly identify those companies that are both financially viable and operationally accessible, allowing them to reach out to potential new vendors more quickly while reducing time spent trying to work with inactive or low-value vendors. The solution addresses the sponsor problem which results in lengthy vendor onboarding procedures and market entry delays.

Methodology Overview (Process Flow)

- Raw vendor data ingestion
- Data cleaning and schema standardization
- Sector and service code classification
- Integration of diversity and performance metrics
- Dashboard visualization and ranking logic
- Strategic decision support and reporting

Vendor data is collected through internal spreadsheets and open procurement platforms. Data cleansing is performed to remove duplicate records; standardize industry classifications; and confirm contact information. After a single data standard (schema) is created, the analysis will filter vendors by sector (i.e., the information and technology sectors). Vendor scores will be created using weighted evaluation metrics for diversity and performance, resulting in an interactive dashboard for executive-level reporting. This process ensures that the analytical process can be repeated, traced, and governed.

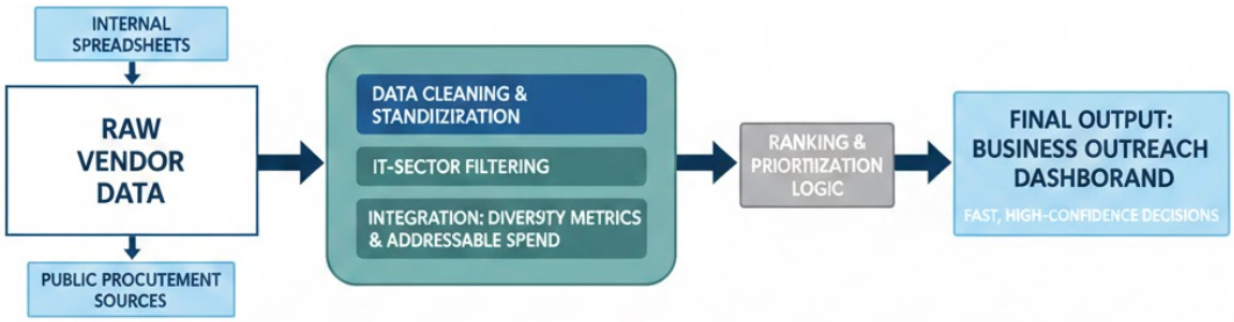


Figure 1.Vendor Prioritization Process Overview.
As shown in the above illustration, the entire process of vendor prioritization begins with raw vendor data. This data is typically ingested through internal spreadsheets and external procurement databases; processed (cleaned up) in order to remove duplicates; filtered to focus only on the appropriate technologies in each vendor; then combine the data from each vendor into one single integrated vendor database; and then create a dashboard presenting the final ranked distribution of vendors to contact for business activity.

Project Timeline and Feasibility (Expanded)

Phase	Duration
Data Collection & Enrichment	3–4 weeks
Cleaning & Standardization	2 weeks
EDA & Weight Calibration	2 weeks
Dashboard Development	3 weeks
Survey Validation	1 week
Refinement	1 week

Total Estimated Timeline: 10–12 weeks

Feasibility is high given structured data sources and limited modeling complexity.

The proposed timeline for completing the project is around 10 to 12 weeks long. The first 3 to 4 weeks will be focused on collecting and augmenting data, followed by approximately 2 weeks on data cleaning and standardizing schemas. An additional 2 weeks will be required to perform exploratory data analysis and calibrate data weights. The consolidated data will then require about 3 weeks of development time to create dashboards and integrate ranking logic; once that development

has been completed, there will be 1 to 2 weeks of stakeholder review and enhancing of the dashboards. The phased process will allow for implementation prior to the end of the academic term.

Risk Considerations and Bias Mitigation (NEW)

Potential risks include:

- Overweighting diversity metrics
- Exclusion bias from verification criteria
- Stakeholder resistance

Mitigation strategies:

- Adjustable weighting
- Documentation of exclusion rules
- Transparent scoring display
- Stakeholder validation

There are several risks associated with these data processes such as the excessive weighting of diversity metrics; the use of strict criteria for vendor contact verification; and stakeholder resistance to formal vendor ranking systems. To counter these risks, weighting of rankings must be flexible; all vendor inclusion criteria must be documented; and validation input must be used to update the scoring algorithms. In addition, this system will include embedded mechanisms for interpretability and executive oversight to reduce unintended bias in the vendor evaluation process.

Conclusion:

The final results of this project demonstrate that the consulting service offered by Summit Global Consulting AI has provided a solution for combining governance, leadership, ethics, and social responsibility in a single analytical platform. This analytical platform will help users interpret fragmented segments of data into cohesive decision-making information to base their purchasing decisions within a healthcare industry upon and align their purchasing decisions with equitable/sustainable business practices.

The purpose of the proposed system is not to predict the outcome with greater accuracy than existing systems but to provide strategic clarity. By establishing an explicit focus on preserving the integrity of the governance and providing interpretability and created structure surrounding how we validate the data, the dashboards will convert a collection of vendor data into a form of operational intelligence that can be utilized to support supplier diversity and executive accountability.

References:

Adobor, H., & McMullen, R. S. (2014). Strategic purchasing and supplier partnerships—The role of a supplier diversity program. *Journal of Business & Industrial Marketing, 29*(4), 263–272. https://doi.org/10.1108/JBIM-03-2012-0051

Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future—Big data, machine learning, and clinical medicine. *The New England Journal of Medicine, 375*(13), 1216–1219. https://www.nejm.org/doi/10.1056/NEJMp1606181

Whitfield, G., & Landeros, R. (2006). Supplier diversity: Portals to prosperity? *Supply Chain Management Review, 10*(5), 16–23. https://www.scmr.com/article/supplier_diversity_portals_to_prosperity

Khatri, V., & Brown, C. V. (2010). Designing data governance. *Communications of the ACM, 53*(1), 148–152. https://doi.org/10.1145/1629175.1629210

Davenport, T. H., & Harris, J. G. (2007). *Competing on analytics: The new science of winning*. Harvard Business School Press